EVERY AFFINE BLOWUP ALGEBRA OF A REDUCED RING IS REDUCED

THE STACKS PROJECT AUTHORS

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Editorial note. The counted statement contains a slogan environment, which the chapter's own preamble defines as a comment environment; the preamble spans of this card start at line 8 and therefore cover it.

Editorial note. The counted spans contain no \ref and no \cite, so the card carries no navigation stub and no bibliography.

Allowed standard prerequisites (editorial).

- Nilpotence in the blowup algebra means a power of a numerator vanishes after clearing denominators. Used at the start of the counted proof (lines 17531-17533). Turns (x/a^n)^m = 0 into a^N x^m = 0 in R.
- Reducedness of R as the absence of nilpotents. Used at the end of the counted proof (lines 17535-17536). Converts (a^e x)^m = 0 into a^e x = 0, which exhibits the original element as zero.

Lemma 0.1. *If R is reduced then every (affine) blowup algebra of R is reduced.*

Proof. Let $I \subset R$ be an ideal and $a \in I$. Suppose x/a^n with $x \in I^n$ is a nilpotent element of $R[\frac{I}{a}]$. Then $(x/a^n)^m = 0$. Hence $a^N x^m = 0$ in R for some $N \geq 0$. After increasing N if necessary we may assume $N = me$ for some $e \geq 0$. Then $(a^e x)^m = 0$ and since R is reduced we find $a^e x = 0$. This means that $x/a^n = 0$ in $R[\frac{I}{a}]$. □

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.